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Case 14-2026: A 50-Year-Old Woman with Vaginal Bleeding and Anemia

Joshua C. Ziperstein, M.D.,[1,2] Rashid A. Barnawi, M.B., B.S.,[3,4] Hanny Al-Samkari, M.D.,[1,2] Marcy B. Bolster, M.D.,[1,2] and Lauren E. Heusinkveld, M.D.[5,6]

PRESENTATION OF CASE

Author affiliations are listed at the end of the article.

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Dr. Tim Niyogusaba (Medicine): A 50-year-old woman was admitted to this hospital because of vaginal bleeding and anemia.

The patient had been in her usual state of health, with leg edema that was attributed to chronic venous insufficiency, until 6 months before the current presentation, when increased swelling and new weakness developed in the legs. Before the onset of these symptoms, she had been physically active. She first noted that her legs felt more tired and sore than usual after normal physical activity, and the symptoms were accompanied by increased swelling and a feeling of heaviness in the legs. Over the course of the next 2 months, the leg weakness progressed to the point that she was unable to rise from a seated position or climb stairs. The patient also noted that her arms felt heavy, and she had difficulty reaching and holding them above her head. During this time, she had intermittent fever that resolved with the use of acetaminophen.

Four months before the current presentation, hair loss developed. What started as patchy hair loss coalesced into larger areas, which prompted the patient to shave her head. When the hair grew back, it was thin and patchy, and she began shaving her head regularly and wearing hats. No scalp pruritus was present, but the patient noted discoloration of the scalp.

Two months before the current presentation, dyspnea on exertion and generalized fatigue developed. The patient stopped working and spent most of her time in her apartment with her legs elevated because of swelling.

One month before the current presentation, the patient was evaluated in the primary care clinic affiliated with this hospital. She had not been seen at the clinic for 2 years. In addition to the symptoms described above, she reported 2 years of irregular vaginal bleeding that occurred almost daily and ranged from spotting to bleeding that required multiple pad changes per day. Physical examination was notable for pitting edema in both legs that extended to the knees, with

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induration and tenderness on palpation of the posterior portion of the upper legs. She had diffuse hair loss and areas of hyperpigmentation across the scalp. She was unable to stand from sitting in a chair without the use of her arms for support. Strength on both sides was 1/5 on knee extension and 3/5 on knee flexion, dorsiflexion, plantar dorsiflexion, elbow flexion, elbow extension, and grip; strength on shoulder abduction, shoulder adduction, and hip flexion was not assessed. A pelvic examination was not performed.

The blood levels of thyrotropin, electrolytes, and glucose were normal, as were results of tests of kidney function. The hemoglobin level was 4.3 g per deciliter (reference range, 12.0 to 16.0), which was a decrease from the baseline level of 8.1 g per deciliter recorded 2 years earlier. The white-cell count was 3370 per microliter (reference range, 4000 to 11,000), the platelet count 123,000 per microliter (reference range 150,000 to 450,000), and the erythrocyte sedimentation rate (ESR) 73 mm per hour (reference range, 0 to 29); other laboratory test results are shown in Table 1. The patient was instructed to present to a local emergency department for evaluation and treatment of anemia.

The patient was admitted to another hospital, where she received a total of 4 units of packed red cells and one infusion of intravenous iron. Her hemoglobin level increased to 8.6 g per deciliter, and the platelet count decreased to 86,000 per microliter. During the admission, ultrasonography of the pelvis showed multiple submucosal uterine fibroids, with the largest measuring 4.5 cm by 4.5 cm by 5.1 cm. No vaginal bleeding was noted during the admission. Her anemia was attributed to blood loss from prolonged vaginal bleeding before admission, and the hair loss was attributed to iron deficiency. Oral iron therapy was initiated, and a follow-up visit in the gynecology clinic was scheduled.

Transthoracic echocardiography, which was performed to evaluate for a cardiac cause of leg edema, showed concentric left ventricular hypertrophy with a normal left ventricular ejection fraction, a mildly dilated right ventricle with mild right ventricular hypertrophy, mild-to-moderate tricuspid regurgitation, and an estimated right ventricular systolic pressure of 29 mm Hg; the main pulmonary artery measured 3.1 cm in diameter. A diagnosis of high-output heart failure leading to pulmonary hypertension was considered. Treatment with furosemide was started, and the leg edema decreased. On hospital day 6, the patient was discharged with plans for follow-up in the cardiology clinic.

After the patient was discharged from the other hospital, the leg edema and dyspnea on exertion resolved, but she continued to have weakness and muscle pain in the arms and legs. She also had persistent difficulty rising from a seated position to standing, climbing stairs, and lifting her arms above her head, along with new difficulty in brushing her teeth because of arm weakness.

Ten days after discharge and 2 weeks before the current presentation, the patient was evaluated at the primary care clinic of this hospital. There was no leg edema. A strength examination was not performed. The complete blood count showed a stable hemoglobin level but worsening thrombocytopenia; other laboratory test results are shown in Table 1. The cause of worsening thrombocytopenia was unknown, and plans were made to complete additional laboratory testing at the time of the patient's next iron infusion.

Two days before the current presentation, the patient noted subjective fever at home. One day before the current presentation, heavy vaginal bleeding developed, with four to five pad changes per hour.

On the day of the current presentation, the patient was evaluated in the primary care clinic of this hospital. She reported ongoing heavy vaginal bleeding with multiple clots, fatigue, and dyspnea on exertion. There was no rash, headache, vision changes, or joint pain. She was brought to the emergency department of this hospital for evaluation and treatment.

On examination, the temporal temperature was 38.1°C, the blood pressure 136/69 mm Hg, the heart rate 98 beats per minute, the respiratory rate 20 breaths per minute, and the oxygen saturation 100% while the patient was breathing ambient air. The body-mass index (the weight in kilograms divided by the square of the height in meters) was 24.7. She had diffuse hair thinning with areas of hypopigmentation and hyperpig-

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Table 1. Laboratory Data.*

Variable	Reference Range, Adults, This Hospital†	1 Mo before Current Presentation	2 Wk before Current Presentation	On Current Presentation
Hemoglobin (g/dl)	12.0–16.0	4.3	9.1	7.2
Hematocrit (%)	36.0–46.0	16.3	31.7	22.2
Mean corpuscular volume (fl)	80.0–100.0	66.8	76.6	78.2
White-cell count (per μl)	4000–11,000	3370	3100	5690
Differential count (per μl)				
Neutrophils	1920–7600	—	—	4430
Lymphocytes	720–4100	—	—	600
Monocytes	160–1100	—	—	600
Eosinophils	0–500	—	—	0
Basophils	0–150	—	—	0
Platelet count (per μl)	150,000–450,000	123,000	52,000	39,000
C-reactive protein (mg/liter)	<8.0	—	—	9.9
Erythrocyte sedimentation rate (mm/hr)	0–29	73	—	88
Aspartate aminotransferase (U/liter)	9–32	49	74	111
Alanine aminotransferase (U/liter)	7–33	31	52	64
Alkaline phosphatase (U/liter)	30–100	48	87	78
Total bilirubin (mg/dl)	0.0–1.0	0.4	1.4	2.5
Direct bilirubin (mg/dl)	0.0–0.3	—	—	0.4
Albumin (g/dl)	3.3–5.0	3.0	3.0	3.2
Total protein (g/dl)	6.0–8.3	8.6	9.1	9.2
Iron (μg/dl)	30–160	17	—	—
Transferrin saturation (%)	14–50	5	—	—
Iron-binding capacity (μg/dl)	230–404	358	—	—
Ferritin (μg/liter)	10–200	10	—	350
Creatine kinase (U/liter)	40–150	—	—	268
Lactate dehydrogenase (U/liter)	110–210	—	—	611
Haptoglobin (mg/dl)	30–200	—	—	<10

* To convert the values for bilirubin to micromoles per liter, multiply by 17.1. To convert the values for iron and iron-binding capacity to micromoles per liter, multiply by 0.1791.
† Reference values are affected by many variables, including the patient population and the laboratory methods used. The ranges used at Massachusetts General Hospital are for adults who are not pregnant and do not have medical conditions that could affect the results. They may therefore not be appropriate for all patients.

mentation involving the entire scalp and bowls of the ears. Cervical lymphadenopathy was present. She had no tenderness on palpation of the abdomen, but splenomegaly was present. The external genitalia were normal. Multiple clots were present in the vagina, and the cervix was deviated posteriorly and to the left. The uterus was mobile on bimanual examination, with an estimated size equivalent to that of a uterus at 10 weeks' gestation. She had no edema in the legs. Strength was 3/5 on shoulder abduction, elbow flexion, and hip flexion and was otherwise 5/5.

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Atrophy of the biceps, triceps, and deltoids was noted.

The patient's medical history included hypertension, migraines, and triphasic Raynaud's phenomenon that had developed 3 years before the current presentation. In addition, she had a history of alcohol use disorder but reported having stopped alcohol consumption 2 years before the current presentation. Medications included ferrous gluconate, furosemide, and polyethylene glycol as needed for constipation. She had no known adverse reactions to medications and no family history of heritable bleeding disorders. The patient worked in sales, and she lived alone in a suburb of Boston. She had smoked approximately five cigarettes per day for the past 20 years and smoked marijuana on most days.

The hemoglobin level was 7.2 g per deciliter, and the platelet count was 39,000 per microliter; other laboratory test results are shown in Table 1. Results of coagulation tests were normal. A peripheral-blood smear showed marked anisopoikilocytosis with numerous hypochromic forms but no schistocytes, spherocytes, or red-cell agglutination. Urinalysis showed 3+ blood (reference value, negative) and 2+ protein (reference value, negative); examination of the urine sediment showed 10 to 20 red cells per high-power field (reference range, 0 to 2). A direct antiglobulin test was negative.

Two units of packed red cells were transfused. Four hours after completion of the second transfusion, the temporal temperature increased to 39.1°C. A sample of blood was obtained for culture, and imaging studies were obtained.

Dr. Rashid A. Barnawi: A chest radiograph (Fig. 1A) showed mild enlargement of the cardiac silhouette, diffuse interstitial opacities, a patchy airspace opacity at the base of the right lung, thickening of the right paratracheal stripe, and increased density in the left mediastinum. Computed tomography (CT) of the chest (Fig. 1B, IC, 1D) and of the abdomen and pelvis (Fig. 1E and 1F), performed after the administration of intravenous contrast material, showed mild bilateral supraclavicular lymphadenopathy and mild-to-moderate bilateral intrathoracic lymphadenopathy; the main pulmonary artery was 3.6 cm in diameter. Splenomegaly was present, with the spleen measuring 18.6 cm in the craniocaudal dimension (reference value, ≤13). She had small-volume ascites and borderline enlarged, multistation abdominopelvic lymphadenopathy.

Dr. Niyogusaba: Vaginal bleeding continued. The patient passed large clots, and the bleeding required multiple pad changes per hour. After transfusion, the hemoglobin level was 7.0 g per deciliter; two additional units of packed red cells were transfused.

The patient was admitted to the hospital. Management decisions were made, and diagnostic tests were performed.

DIFFERENTIAL DIAGNOSIS

Dr. Joshua C. Ziperstein: This 50-year-old woman presented with acute and life-threatening vaginal bleeding for which she received emergency supportive care. However, determining the correct diagnosis to explain her presentation requires consideration of her progressive signs and symptoms over the preceding weeks, months, and years. In developing a differential diagnosis, I will begin by grouping her signs and symptoms into three phases of illness: immune initiation, immune activation, and immune culmination.

IMMUNE INITIATION

The first phase of this patient's illness started with the development of Raynaud's phenomenon 3 years before the current presentation, followed by irregular vaginal bleeding associated with uterine fibroids and systemic symptoms that include proximal muscle weakness, leg edema, fever, alopecia, dyspnea on exertion, and fatigue. I would label this first phase of illness as immune initiation, since I suspect that the patient's systemic symptoms are connected pathophysiologically to Raynaud's phenomenon and possibly to bleeding fibroids. Endometriosis may represent a unifying diagnosis for these syndromes, given the growing evidence of epidemiologic and biologic links between endometriosis and fibroids[1] and the increasing observational[2] and genetic[3] links between endometriosis and autoimmune phenomena. Although these pathophysiological associations are interesting, the lack of data on a platelet trend during this period of time makes it hard to establish a mechanistic link between her irregular vaginal bleeding and her immune system.

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IMMUNE ACTIVATION

The second phase of this patient's illness is characterized by signs that point to more substantial activation of the immune system — leukopenia, thrombocytopenia, an elevated ESR, an elevated globulin gap, and cardiac ultrasound findings from another hospital that showed increased filling pressures in the right side of the heart.

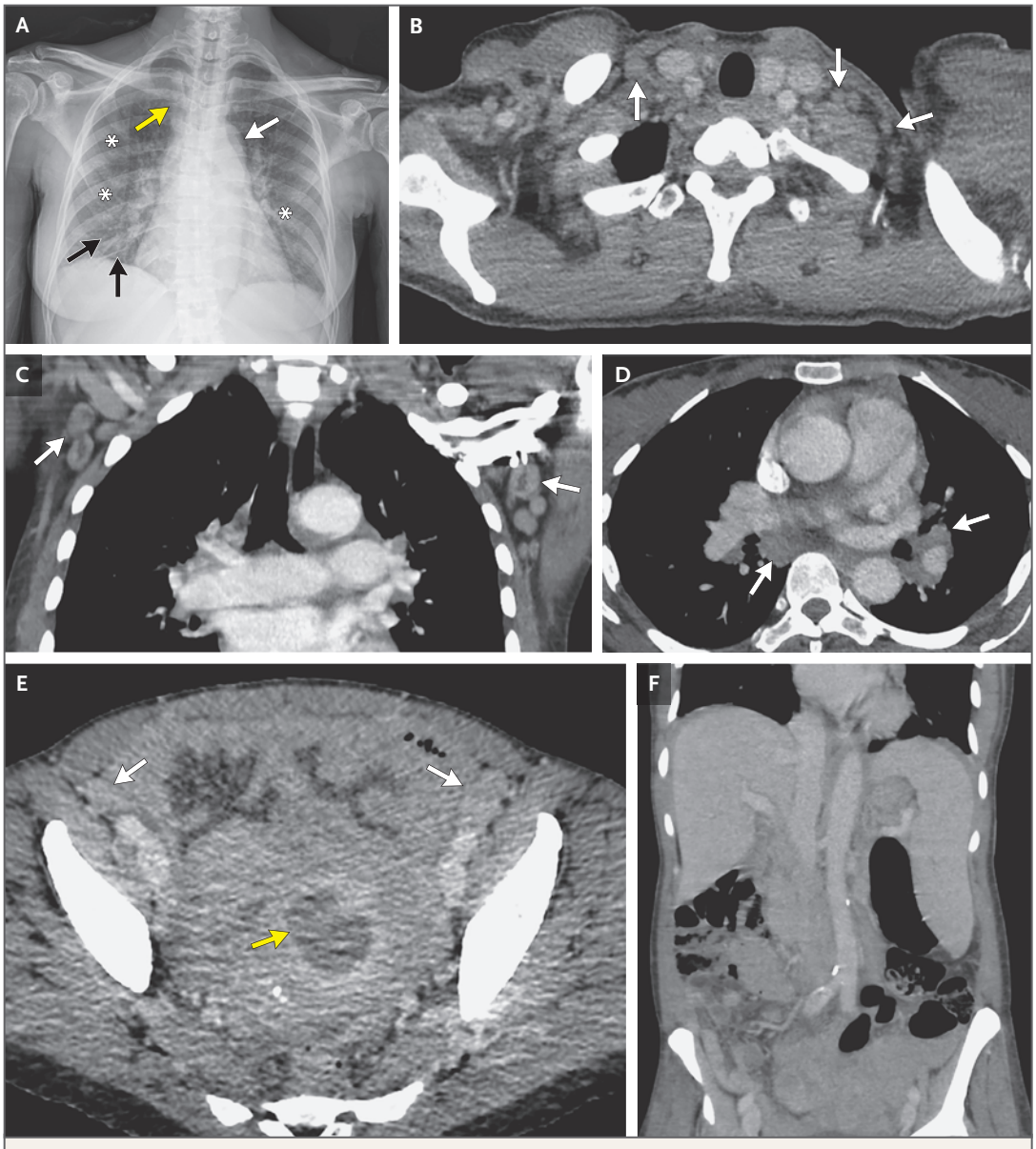


Figure 1. Imaging Studies.

A radiograph of the chest obtained in a frontal view (Panel A) shows diffuse interstitial opacities (asterisks), a patchy airspace opacity at the base of the right lung (black arrows), thickening of the right paratracheal stripe (yellow arrow), and increased density in the left mediastinum (white arrow). CT of the chest, abdomen, and pelvis was performed after the administration of intravenous contrast material (Panels B through F). An axial image of the upper chest (Panel B) shows bilateral supraclavicular lymphadenopathy (arrows). A coronal image of the chest (Panel C) shows bilateral axillary lymphadenopathy (arrows), and an axial image (Panel D) shows mediastinal and hilar lymphadenopathy (arrows). An axial image of the pelvis (Panel E) shows bilateral external iliac (pelvic) lymphadenopathy (white arrows) and a hypoattenuating uterine lesion consistent with a fibroid (yellow arrow). A coronal image of the abdomen and pelvis (Panel F) shows splenomegaly.

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I suspect that the lymphadenopathy, splenomegaly, and proteinuria that were eventually discovered on admission to this hospital were most likely also present during this phase.

IMMUNE CULMINATION

The third phase of this patient's illness is marked by an emergency presentation to this hospital with fibroid-related hemorrhage requiring aggressive transfusion support. I suspect that this patient's bleeding is most likely a consequence of profound thrombocytopenia. This phase of illness can be described as the culmination of ongoing and escalating immune system activation, as evidenced by fever, lymphadenopathy, further elevation of the ESR and globulin gap, and progressive thrombocytopenia.

After organizing this patient's collection of progressive signs and symptoms into a linear and descriptively framed illness schema of immune initiation, activation, and culmination, the next step is to focus on disease paradigms that could match this progression. When evaluating patients with immune activation, I consider three categories of disease: dysfunctional hematopoiesis (i.e., cancer), dysfunctional antigen response (i.e., autoimmunity), and functional antigen response (i.e., infection).

The relatively indolent initial course of this patient's illness is suggestive of an autoimmune disease, as is the development of a broader immune syndrome associated with Raynaud's phenomenon. Although her constitutional symptoms and hematologic abnormalities could also be consistent with cancer or infection, the muscle weakness, alopecia, and cardiac hemodynamic findings suggestive of an interstitial lung process are more consistent with an autoimmune disease. The discordance in the degree of elevation between the ESR and the the C-reactive protein level is notable. However, this pattern does not have statistically significant support for an autoimmune diagnosis.[4]

SYSTEMIC LUPUS ERYTHEMATOSUS

Within the autoimmune category, this patient's syndrome is most consistent with a diagnosis of systemic lupus erythematosus (SLE). Although it is challenging to parse specific autoimmune syndromes without the results of autoantibody serologic testing, the features of this patient's case that best fit with a diagnosis of SLE include hematologic abnormalities, multisystem involvement, escalating clinical acuity, the absence of joint symptoms, and the absence of pathognomonic features of another autoimmune disease. Finally, I will return to the data showing discordance between the ESR and the C-reactive protein level. Although a discordantly elevated ESR is not statistically predictive of autoimmune disease in general, it does track closely with a diagnosis of SLE.[4]

However, three clinical features of this patient's presentation would give me pause in making a diagnosis of SLE, since they are rare findings in patients with this condition. First, she has extensive multistation lymphadenopathy, which seemed to be profound as compared with historical descriptions of lymphadenopathy in patients with SLE. This finding is of particular interest because infection and cancer are possible diagnoses in this case.

Second, an elevated lactate dehydrogenase (LDH) level and undetectable haptoglobin level are findings suggestive of hemolytic anemia. A microangiopathic process, which is rare in SLE, could be suggestive of an alternative diagnosis. However, a detailed review of the peripheral-blood smear showed no schistocytes, which essentially ruled out a microangiopathic hemolytic anemia syndrome. I presume that the LDH and haptoglobin findings are related to a nonmicroangiopathic hemolytic process, which is more common than a microangiopathic hemolytic process in SLE; alternatively, these laboratory findings could be explained by her acute bleeding, massive transfusion, or both.

Third, this patient had clinical findings (weakness and muscle atrophy) and laboratory findings (elevated levels of creatine kinase and aspartate aminotransferase) that showed evidence of not just myalgia but also myositis. Although myalgia commonly occurs in patients with SLE, true myositis might suggest an alternative diagnosis. I was somewhat reassured to learn that inflammatory muscle disease in SLE can be variable and that the level of creatine kinase in this patient was on the lower end of the typical range in patients with SLE who have a concurrent diagnosis of inflammatory myositis.[5]

EXPLAINING ESCALATING CLINICAL ACUITY

One loose end in considering a diagnosis of SLE is the need to explain why this patient's clinical

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status worsened so quickly and profoundly. Could SLE have been provoked by the recent administration of intravenous iron[6] or packed red cells[7] at the other hospital? Alternatively, could SLE have been an immune risk factor for the development of another hematologic process, such as lymphoma or multicentric Castleman disease? During the admission to this hospital, she had a possible febrile transfusion reaction. This event reinforces my theory that she has an autoimmune disease that could react to red-cell transfusion, which supports the explanatory model of a provoking factor that shifted SLE from the "immune activation" to the "immune culmination" phase of illness. However, it is possible that autoimmune disease was a risk factor for the development of cancer, or that cancer explains her entire course of illness.

I suspect that this patient has SLE. Therefore, my next diagnostic step would be to rapidly and noninvasively evaluate for this disease with serologic testing for autoantibodies and to concurrently assess for a hematologic cancer with the use of flow cytometry. If these tests do not provide enough diagnostic confidence to begin treatment, I would plan to administer packed red cells and platelets and move forward with prompt tissue sampling.

DR. JOSHUA C. ZIPERSTEIN'S DIAGNOSIS

Systemic lupus erythematosus.

INITIAL DIAGNOSTIC TESTING

Dr. Lauren E. Heusinkveld: An immunofluorescence assay performed on Hep-2 (human epithelial) cells showed antinuclear antibodies (ANA) with a homogeneous pattern (titer, 1:640) and ANA with a speckled cytoplasmic pattern (titer, 1:80). Specific antibody testing was positive for anti-Smith antibodies, anti-ribonucleoprotein (anti-RNP) antibodies, anti–cyclic citrullinated protein, and rheumatoid factor and was equivocal for antibodies against Sjögren's syndrome–related antigen A (anti-Ro antibodies). Antibodies against Sjögren's syndrome–related antigen B (anti-La antibodies) and anti–double-stranded DNA antibodies were not detected, and an autoimmune inflammatory myopathy IgG panel was negative. A skin punch-biopsy specimen, which was obtained from the left postauricular area of the scalp (Fig. 2), showed marked alopecia with miniaturization of the few remaining hair follicles, perifollicular and epidermal interface dermatitis with vacuolar changes and pigment incontinence, and markedly increased deposition of dermal mucin. The histologic findings of interface dermatitis combined with dermal mucin deposition help to establish the diagnosis of a systemic autoimmune rheumatic disease. The positive serologic tests taken together with the skin-biopsy findings support the diagnosis of SLE.

HOSPITAL COURSE

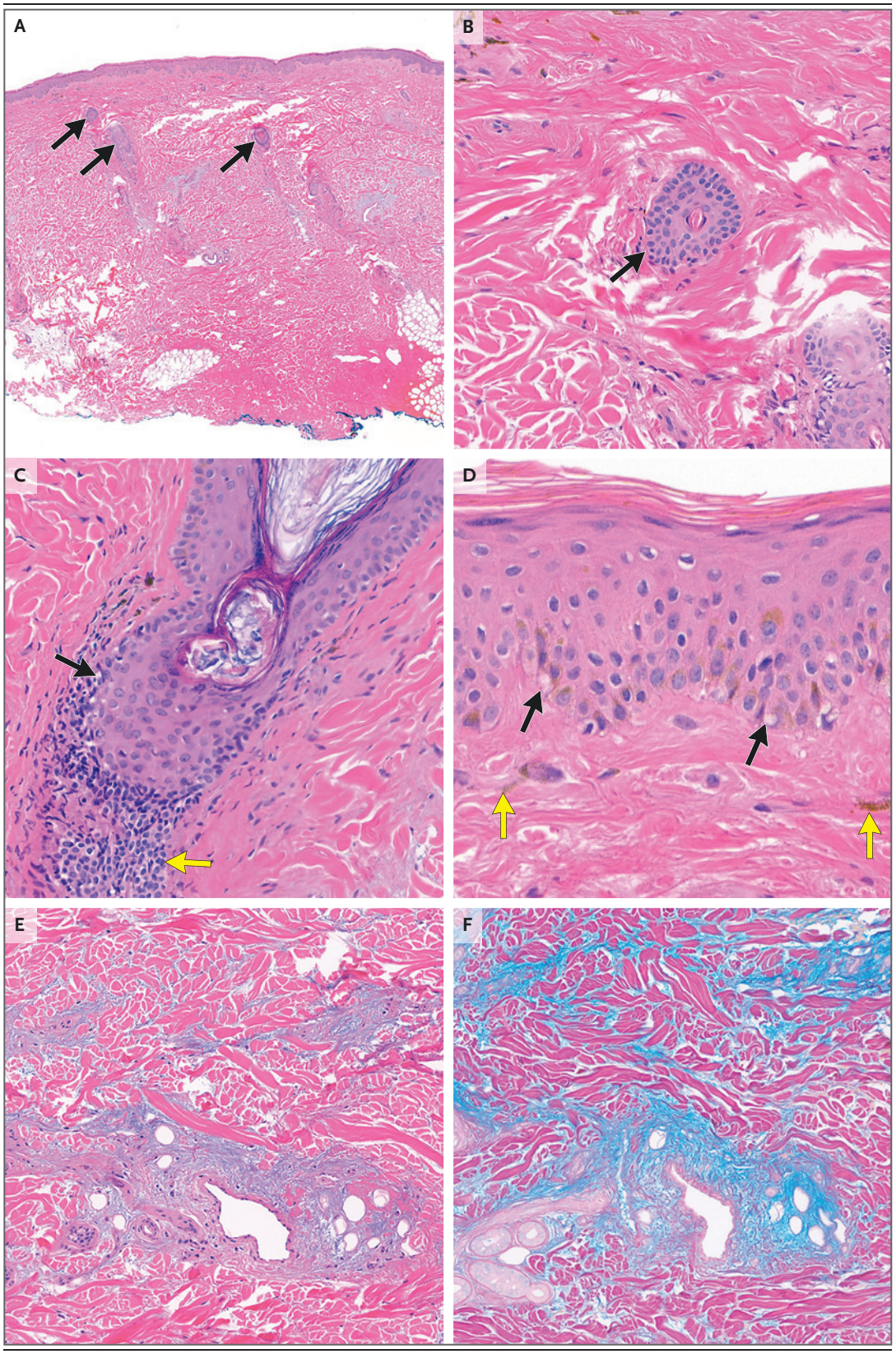
Dr. Niyogusaba: Despite treatment with oral tranexamic acid and the transfusion of one pack of platelets, vaginal bleeding persisted, and five additional units of packed red cells were transfused, with no increase in the level of hemoglobin. On hospital day 3, the patient underwent laparoscopic total hysterectomy and bilateral salpingectomy, which revealed a uterus with numerous pedunculated fibroids, including one submucosal prolapsing fibroid measuring 5 cm in diameter. Examination of the surgical specimen showed leiomyomata and an inactive endometrium.

After the surgical procedure, the hemoglobin level stabilized without the need for further transfusion of packed red cells; however, thrombocytopenia persisted. A direct assay for platelet-associated IgG antibodies was negative, and a

Figure 2 (facing page). Biopsy Specimen of the Left Postauricular Scalp.

Hematoxylin and eosin staining of a skin punch-biopsy specimen obtained from the left postauricular scalp shows alopecia (Panel A), along with miniaturization of the hair follicles (Panel B). The few remaining hair follicles are markedly reduced in size (Panels A and B, arrows). Perifollicular interface dermatitis (Panel C) is present with a lymphocytic inflammatory infiltrate (yellow arrow) and vacuolar changes of the follicular epithelium (black arrow). Epidermal interface dermatitis (Panel D) is present with vacuolar changes (black arrows) and dermal pigment incontinence (yellow arrows) with sparse inflammation. Markedly increased deposition of dermal mucin is present, as evidenced by the grayish-blue coloration pooling around vessels and interspersed between dermal collagen bundles (Panel E). Colloidal iron staining (Panel F) confirms the increased dermal mucin deposition (in blue).

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direct assay for platelet-associated IgM antibodies was strongly positive.

The urinary protein:creatinine ratio was 6.51 (reference value, <0.15), and the blood level of creatinine remained normal. On hospital day 14, a biopsy of the kidney was performed.

PATHOLOGICAL DISCUSSION

Dr. Heusinkveld: A core-needle biopsy specimen was obtained from the left kidney (Fig. 3). Light microscopy showed mild endocapillary hypercellularity in 3 of 18 glomeruli with membranous deposits. Immunofluorescence staining was positive for IgG, IgM, IgA, C3, and C1q (a finding known as a "full house" pattern) and for kappa and lambda immunoglobulin light chains, with predominantly membranous IgG and C3 deposition, a finding that is diagnostic of lupus nephritis in the context of positive serologic tests. Electron microscopy showed mesangial and subepithelial deposits and tubuloreticular structures, findings that are associated with systemic interferon activity and are common in patients with lupus nephritis. A classification of lupus nephritis is based on glomerular morphologic features and the distribution of immune complex deposits.

PATHOLOGICAL DIAGNOSIS

Lupus nephritis, predominantly class V (membranous), with a mild class III (endocapillary hypercellularity) component.

DISCUSSION OF MANAGEMENT

Dr. Hanny Al-Samkari: The patient's severe thrombocytopenia in the context of a presentation consistent with a systemic autoimmune disease suggested possible secondary immune thrombocytopenia (ITP). Given this possibility, testing for platelet-associated antibodies was performed, which was negative for platelet-associated IgG but strongly positive for platelet-associated IgM.

In the diagnosis of ITP, platelet autoantibody testing is controversial. However, it is a useful tool in ambiguous cases such as this one, given that misdiagnosis of ITP is common, even among expert classical hematologists.[8,9] The clinical performance of platelet autoantibody testing is highly dependent on numerous analytical and preanalytical factors, including the assay methods used.[10] The International Society on Thrombosis and Haemostasis guidelines recommend the use of direct tests for platelet autoantibodies, which measure antibodies on the platelet surface, rather than indirect tests, which measure antibodies in plasma. The guidelines also recommend glycoprotein-specific tests that assess for antibodies specific to glycoprotein IIb–IIIa and glycoprotein Ib–IX.[11] The initial platelet-associated antibody test performed on a sample of this patient's blood was a direct test, but it did not assess for glycoprotein-specific antibodies. Circulating antibodies that are nonspecific for platelet surface antigens commonly bind to platelets, particularly in inflammatory states when antibody titers are elevated, which may lead to false positive results on platelet-associated antibody tests that are not glycoprotein-specific.

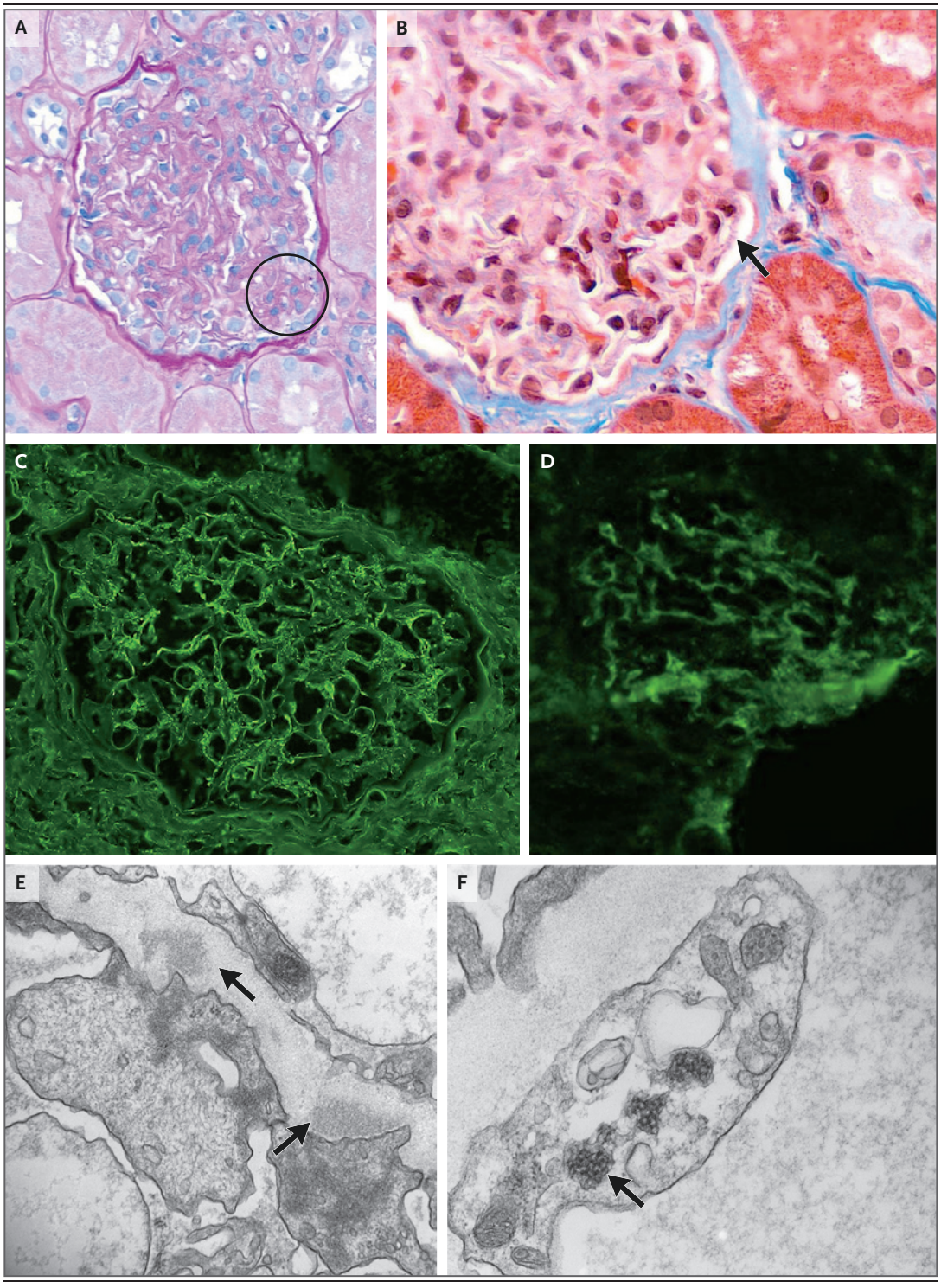
A subsequent direct glycoprotein-specific platelet autoantibody test that was performed on a sample of this patient's blood was negative. This finding indicates that the initial strongly positive test for platelet-associated IgM was a false positive result, most likely caused by the high titer of rheumatoid factor IgM antibodies. This negative result, along with the normalization of the platelet count, which coincided with the decrease in vaginal bleeding and resolution of splenomegaly, confirmed that the predominant cause of the patient's thrombocytopenia was not immune-mediated destruction. Rather, it was multifactorial in the context of persistent

Figure 3 (facing page). Biopsy Specimen of the Kidney.
Periodic acid–Schiff staining (Panel A) shows mild endocapillary proliferation, identified in 3 glomeruli (18%) (circle). Trichrome staining (Panel B) shows membranous glomerular deposits (arrow). Immunofluorescence staining shows focal and segmental, predominantly membranous deposits in the glomerular basement membrane, which are positive for IgG (Panel C) and C1q (Panel D), as well as C3, IgA, IgM, and kappa and lambda light chains (not shown). Images from electron microscopy (Panels E and F) show subepithelial deposits (Panel E, arrows) and mesangial deposits (not shown), as well as numerous tubuloreticular structures (Panel F, arrow), which are associated with systemic interferon activity and are commonly seen in patients with lupus nephritis.

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platelet consumption from chronic hemorrhage and splenic sequestration due to splenomegaly.

Dr. Marcy B. Bolster: A rheumatology consultation was requested early in this patient's hospital course. The rash on the scalp and ears was consistent with discoid rash, and hair loss was consistent with patchy alopecia. Proximal muscle weakness was present on both sides; strength



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was 4/5 on flexion of the hips and neck and in the deltoids and triceps. Electromyography revealed changes suggestive of an inflammatory myopathy. The patient declined both magnetic resonance imaging of the muscle and a muscle biopsy.

After a review of the history, physical examination findings, and objective test results, this patient received a diagnosis of an overlap syndrome of SLE and idiopathic inflammatory myopathy. The presence of anti-RNP antibodies and features consistent with several autoimmune diseases led to consideration of mixed connective-tissue disease, a disorder characterized by positive anti-RNP antibodies and features of SLE, idiopathic inflammatory myopathy, systemic sclerosis, and rheumatoid arthritis. However, because classification criteria were fulfilled[12,13] for two rheumatic diseases, the more precise diagnosis in this patient is overlap syndrome of SLE and idiopathic inflammatory myopathy; treatment was appropriately directed at the visceral manifestations of these two conditions.

Even before the results of the kidney biopsy became available, treatment with hydroxychloroquine was initiated for SLE on the basis of its benefits in reducing flares, limiting organ damage, and improving survival.[14-16] In patients with idiopathic inflammatory myopathy, treatment with glucocorticoids would usually be considered. However, because of ongoing evaluation of multistation lymphadenopathy and splenomegaly, the patient instead received intravenous immune globulin, after which muscle strength quickly improved to 4+/5 in all proximal muscle groups.

A lymph-node biopsy specimen showed granulomatous changes without evidence of cancer. Treatment for lupus nephritis was initiated with a triple therapy approach, which consisted of glucocorticoids, mycophenolate mofetil, and plans for outpatient belimumab initiation.[17] Treatment with trimethoprim–sulfamethoxazole was started for prophylaxis against *Pneumocystis jirovecii* pneumonia.

Belimumab was added to the patient's treatment regimen 2 months after hospital discharge. She has maintained close follow-up, and her creatine kinase levels have normalized. The creatinine levels have remained normal, and proteinuria has resolved. She has had complete normalization of proximal muscle strength, resolution of alopecia, and an increase in energy.

FINAL DIAGNOSIS

Overlap syndrome of systemic lupus erythematosus and idiopathic inflammatory myopathy.

This case was presented at the Medicine Case Conference.

Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

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AUTHOR INFORMATION

[1]Department of Medicine, Massachusetts General Hospital, Boston; [2]Department of Medicine, Harvard Medical School, Boston; [3]Department of Radiology, Massachusetts General Hospital, Boston; [4]Department of Radiology, Harvard Medical School, Boston; [5]Department of Pathology, Massachusetts General Hospital, Boston; [6]Department of Pathology, Harvard Medical School, Boston.

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